Luis Miguel Loaiza Zapata

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Profile

I am a mechanical engineer with a strong interest in computational mechanics, robotics, and biomedical engineering. My experience spans CAD design (SolidWorks), scientific programming (Python, MATLAB, C++), and simulation tools such as ANSYS, alongside applied research in areas like predictive maintenance, medical image processing, and haptic sensing. I am an independent, proactive person who is passionate about continuous learning and problem-solving.

Education

Erasmus Mundus Joint Master's Programme, M.Sc. in Biomedical Engineering — Serbia, Greece and Romania
University of Kragujevac **- Serbia** — 2025 – Present
University of Patras **- Greece**
Grigore T. Popa University of Medicine and Pharmacy Iași **- Romania**

EAFIT University, B.S. in Mechanical Engineering — Medellin, Colombia
Jan 2020 – Dec 2024

- GPA: 4.6 / 5.0 (Scale 0.0 – 5.0)
- Graduated as Rank 1 in the Mechanical Engineering program

Experience

Research Intern, Open Robotics Group, University of Patras — Patras, Greece
Jun 2026 – Sep 2026

- Conducted research on haptic feedback technologies for prosthetic devices.
- Investigated the development of electronic skins (e-skins) for tactile sensing applications.

Project Engineer, ALICO S.A.S | BIC — Medellin, Colombia
May 2025 – Sep 2025

- Designed mechanical components for industrial machines and produced technical drawings for manufacturing.
- Developed software tools and applications to optimize internal tasks.
- Evaluated technologies for predictive maintenance applications.

Mechanical Engineering Intern, ISAGEN S.A. — Medellin, Colombia
Jan 2024 – Jun 2024

- Conducted research for the project *"Automated diagnosis of machines using process and predictive variables."*
- Evaluated machine learning–based technologies for predictive maintenance applications.
- Analyzed mechanical vibrations in power generation units in collaboration with the digital transformation team.

Key Projects

CFD Simulation in Coronary Artery for Blood Flow Analysis, University of Patras — Patras, Greece
Feb 2026 – Jun 2026

- Reconstructed a patient-specific coronary artery geometry using specialized reconstruction software.
- Performed haemodynamic analysis using ANSYS to characterize blood flow behavior within the artery.
- Analyzed flow patterns relevant to the detection of coronary artery disease.
- Tools Used: ANSYS, CFD

3D Rib Cage Segmentation, University of Kragujevac — Kragujevac, Serbia
Nov 2025 – Jan 2026
github.com/repo

- Developed three segmentation approaches for rib structures using two publicly available datasets: RibFrac (CT-based rib instance and fracture segmentation) and VinDr-RibCXR (chest X-ray rib labeling).
- Adapted and integrated two public repositories to perform rib fracture detection and individual rib segmentation and labeling.
- Tools Used: Python, Deep Learning frameworks

Integration of the multipole method to BEM method, Eafit — Medellin, Colombia
July 2024 – Nov 2024
github.com/repo

- Adapted a boundary element code base (BEM) to solve Poisson equation problems.
- Implemented multipoles to reduce computational overhead and improve algorithm efficiency.
- Tools Used: Python, Gmsh

Robotic Arm programming using ROS2 system, Robotics Research Group ARES — Medellin, Colombia
July 2023 – Nov 2023
github.com/repo

- Learned basic ROS2 concepts and compatible tools
- Programmed robotic movement in virtual and real environments.
- Tools Used: C++, Python, ROS2, Linux, Rviz, Gazebo

Certifications

- Certified SOLIDWORKS Simulation and Mechanical Design Associate — Sept 2024
- Certified Foundational MATLAB - *MathWorks* — July 2024
- Certified Python in Data Science *DataCamp* — March 2024

Achievements

- Recipient of the *Erasmus Mundus* Scholarship for the M.Sc. in Biomedical Engineering EMMBIOME. — June 2025
- Recipient of the *COLFUTURO* Scholarship for graduate studies abroad. — May 2025
- Winner of the Mechanical Engineering Inventive Project Award. — Oct 2021
- Recipient of the *Fundación Educación* Scholarship for undergraduate studies. — Dec 2019

Skills

Languages: Spanish (Native), English (C1 - IELTS 7.0), German (Basic - A2)

Programming Languages: Python, Matlab, R, C++

Technologies: SolidWorks, Matlab, VSCode, Microsoft Office

References

Dimitar Stanev - Senior Consultant - Capgemini Engineering

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Antonios Sakellarios - Assistant Professor- University of Patras

- **Email:** asakellarios@upatras.gr

Konstantinos Moustakas - Professor- University of Patras

- **Email:** moustakas@upatras.gr